Project Title	**Uber Eats Bangalore Restaurant Intelligence & Decision Support Systems**
Skills take away From This Project	**Python, Pandas, MySQL, Streamlit, Statistics, Data Organizing, Data Cleaning.**
Domain	**SQL / Data Engineering / Python/ Data Scientist / Data Analyst**

Problem Statement

Uber Eats operates a large-scale restaurant marketplace where business success depends on factors such as location strategy, pricing, cuisine mix, customer ratings, and platform features like online ordering and table booking.

The objective of this project was to analyze Uber Eats Bangalore restaurant data and build a decision support system that answers critical business questions using Python and SQL, presenting results as clean tabular DataFrame outputs in Streamlit (no charts or visualizations).

This approach mirrors real internal analytics dashboards, where stakeholders often require precise tabular insights rather than visual storytelling.

Business Use Cases

- Location Intelligence
- Partner Onboarding Strategy
- Pricing Optimization

- Cuisine Performance Analysis
- Product Feature Impact Assessment
- Market Segmentation
- Customer Satisfaction Drivers
- Expansion & Premium Restaurant Planning

Approach

Data Extraction & Transformation (Python)

- Source data loaded from CSV
- Data cleaning using Pandas:
  - Duplicate removal
  - Missing value handling
  - Rating normalization
  - Cost standardization
  - Feature engineering for pricing segments and rating categories

Database Layer (SQL)

- Cleaned data stored in Relational Database - example- MySQL
- Cursor-based SQL queries executed
- Heavy use of:
  - GROUP BY, HAVING
  - CASE WHEN for segmentation

3.Streamlit Application (No Visualization)

The Streamlit application is designed without graphical visualizations and is used exclusively for:

- Executing SQL queries
- Displaying DataFrame-based results
- Allowing business users to explore insights through structured tabular outputs

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Application Structure

1. Dashboard Page

- Provides multiple filtering options for users
- Allows users to filter and retrieve data dynamically from the SQL database
- Displays the filtered results in DataFrame/table format
- All filtering logic is implemented strictly using SQL queries

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2. Q&A Page

- Displays analysis for 10 predefined business questions
- Each answer is generated through SQL-based computation
- Results are presented in structured table format within Streamlit

3. Order Data Integration & Custom Analytical Q&A

In addition to existing datasets:

- The generated order dataset (originally stored in JSON format) will be imported and stored in the SQL database as a structured table.
- All order-related analysis will also be executed using SQL queries.
- A custom Q&A module will be implemented specifically for the order dataset
- The system will support answering as many meaningful business queries as possible, leveraging SQL-based computation for consistency and efficiency

Key Business Questions Answered (Any 10 Questions)

1. Which Bangalore locations have the highest average restaurant ratings?
   Business Value: Identifies premium-performing areas suitable for brand positioning and new partner onboarding.

2. Which locations are over-saturated with restaurants?
   Business Value: Helps avoid overcrowded markets and guides smarter expansion decisions.

3. Does online ordering improve restaurant ratings?
   Business Value: Evaluates the ROI of Uber Eats' online ordering feature for partners.

4. Does table booking correlate with higher customer ratings?
   Business Value: Measures the effectiveness of table booking as a premium feature.

5. What price range delivers the best customer satisfaction?
   Business Value: Helps define the optimal pricing segment for partner success.

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6. How do low, mid, and premium-priced restaurants perform in terms of ratings?
   Business Value: Supports pricing-based market segmentation strategies.

7. Which cuisines are most common in Bangalore?
   Business Value: Reveals market demand and cuisine saturation levels.

8. Which cuisines receive the highest average ratings?
   Business Value: Identifies high-quality cuisine categories suitable for promotion.

9. Which cuisines perform well despite having fewer restaurants?
   Business Value: Highlights niche opportunities for differentiation.

10. What is the relationship between restaurant cost and rating?
    Business Value: Determines whether higher pricing translates to better customer perception.

11. Which locations are ideal for premium restaurant onboarding?
    Business Value: Combines cost, rating, and location insights to guide premium expansion.

12. Which locations show high demand but lower average ratings?
    Business Value: Indicates areas where quality improvement initiatives are needed.

13. Do restaurants offering both online ordering and table booking perform better?
    Business Value: Validates bundled feature adoption for partners.

14. What combination of factors maximizes restaurant success on Uber Eats?
    (Pricing + Location + Cuisine + Platform Features)
    Business Value: Supports strategic partner recommendations.

15. Which restaurants are top performers within each pricing segment?
    Business Value: Helps identify benchmark partners and best practices.

Results

- Identified top-performing and over-saturated locations in Bangalore
- Confirmed that mid-priced restaurants achieve the highest average ratings
- Demonstrated that online ordering and table booking positively impact ratings
- Uncovered cuisine-specific performance and niche opportunities
- Delivered a pure tabular, decision-focused Streamlit application

Project Evaluation Metrics

- Clean and maintainable Python code
- Modular scripts for scalability
- SQL-driven analytics (no hardcoding)
- Streamlit-based DataFrame outputs only
- Strong alignment with real business decision-making
- GitHub repository with clear documentation

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Technical Stack

- **Language:** Python
- **Database:** SQLite / MySQL (cursor-based SQL)
- **Application:** Streamlit
- **Libraries:**
    - Pandas
    - NumPy
    - sqlite3 / mysql-connector-python

Dataset

- **Source:** Uber Eats Bangalore Restaurants Dataset: 1)**Dataset**, 2) **Order_json**
- **Key Columns:**
    - restaurant_name
    - location
    - cuisines
    - rate
    - approx_cost_for_two
    - online_order
    - book_table

Project Deliverables

- SQL database with cleaned restaurant data
- Python scripts for:
    - Data cleaning & preprocessing

    - SQL insertion and querying
- Streamlit application displaying:
    - 15 business questions
    - DataFrame-based answers
- Well-documented README file

Timeline

- **Project Duration:** 14 Days from assigned date

References:

In github, upload all your code files and submit the repository link in the Project section of your ZEN portal and you also have to appear for Project evaluation with one of our Mentors.

*****If you don't know how to approach the project, kindly refer to the project orientation recording provided in this table.***

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Project Live Evaluation	Project Live Evaluation
EDA Guide	Exploratory Data Analysis (EDA) G…
Capstone Explanation Guideline	Capstone Explanation Guideline
GitHub Reference	How to Use GitHub.pptx
Project_Series_English	Project_Series_English

Project_Series_Tamil	Project_Series_Tamil
Streamlit Doc	Streamlit doc
Streamlit Reference English	Streamlit Recordings
Streamlit Reference Tamil	Streamlit Recording Tamil
Project Orientation (English)	Project Orientation Session : Uber …
Project Orientation (Tamil)	Uber Eats Bangalore Restaurant Int…
SQL_Lite	SQL_Lite
Vs code	Vs code

PROJECT DOUBT CLARIFICATION SESSION (PROJECT AND CLASS DOUBTS)

About Session: The Project Doubt Clarification Session is a helpful resource for resolving questions and concerns about projects and class topics. It provides support in understanding project requirements, addressing code issues, and clarifying class concepts. The session aims to enhance comprehension and provide guidance to overcome challenges effectively.

Timing: Monday-Saturday (3:00PM to 4:00PM)

link :PROJECT DOUBT CLARIFICATION SESSION

LIVE EVALUATION SESSION (CAPSTONE AND FINAL PROJECT)

About Session: The Live Evaluation Session for Capstone and Final Projects allows participants to showcase their projects and receive real-time feedback for improvement. It assesses project quality and provides an opportunity for discussion and evaluation.
Note: This form will Open only on Saturday (after 2 PM) and Sunday on Every Week

Timing: Monday-Saturday (05:30PM to 07:00PM)

Booking link : https://forms.gle/1m2Gsro41fLtZurRA

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